

The smallest singular value of a sparse Mertens matrix

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July 25, 2026

Abstract

Redheffer's $(0, 1)$ matrix R_n satisfies $\det R_n = M(n) := \sum_{k \leq n} \mu(k)$, and the Riemann hypothesis is equivalent to $M(n) = O(n^{1/2+\varepsilon})$. A substantially sparser $(0, 1)$ matrix $\mathcal{R}_n$ with the same determinant — about $2.61n$ nonzero entries against Redheffer's $\sim n \log n$ — was introduced in [13], and in [14] its two dominant *eigenvalues* were shown to be $1 \pm \sqrt{\pi(n)} + O(\log \log n)$. We determine its *singular values* at both ends. The largest is $\sigma_{\max}(\mathcal{R}_n) = \sqrt{n}(1 + O(1/n))$, forced by a single dense row, with non-normality gap $\sigma_{\max}/|\lambda_{\max}| \asymp \sqrt{\log n}$. The smallest satisfies

$$|M(n)| \cdot n^{-3/2+o(1)} \leq \sigma_{\min}(\mathcal{R}_n) \leq |M(n)| \cdot n^{-4/3+o(1)},$$

and, established separately, $\text{RH} \iff \sigma_{\min}(\mathcal{R}_n) \ll_{\varepsilon} n^{-1+\varepsilon}$, two-sided and unconditional. The forward implication does not use the displayed upper bound, which under RH would give only $n^{-5/6+\varepsilon}$; it uses the collapsed equality $\sigma_{\min}(\mathcal{R}_n) = |M(n)|n^{-3/2+o(1)}$, valid as soon as $|M(n)|/\|w\| \rightarrow 0$, which RH itself supplies. The mechanism is that the entries of the vector $\mathcal{A}^{-\top}(\mathbf{1} - e_1)$, where $\mathcal{A}$ is the unit lower-triangular factor of $\mathcal{R}_n$, are *exactly* the sums $M(x, y) = \sum_{d \leq x, P^-(d) > y} \mu(d)$ of the Möbius function over integers free of small prime factors; the analytic input is then Alladi's asymptotic $M(x, x^{1/\alpha}) \sim x\rho'(\alpha)/\log y$ with ρ Dickman's function [1], which we cite rather than claim. It gives $\|w\| = n^{1+o(1)}$ unconditionally, and that norm is what produces the exponent $-3/2$ above. Along the way we show $\|\mathcal{A}^{-1}\| = n^{1/2+o(1)}$ via a Neumann series that terminates after $\sim \log n / \log \log n$ terms, hence $\sigma_{\min}(\mathcal{A}) = n^{-1/2+o(1)}$; this answers, negatively for this family, a question of Cheon and Kim [8]. We also record why the problem is severely pre-asymptotic: several natural conjectures suggested by computation up to $n = 10^7$ are false, and we identify the finite-range mechanism responsible.

1 Introduction

Write μ for the Möbius function and

$$M(n) := \sum_{k \leq n} \mu(k)$$

for the Mertens function. It is classical that the prime number theorem is equivalent to $M(n) = o(n)$, and that the Riemann hypothesis is equivalent to

$$M(n) = O(n^{1/2+\varepsilon}) \quad \text{for every } \varepsilon > 0. \quad (1)$$

Redheffer [17] introduced the $n \times n$ $(0, 1)$ matrix R_n with $R(i, j) = 1$ when $j = 1$ or $i \mid j$, and proved $\det R_n = M(n)$. This places (1) inside linear algebra, and a long line of work has studied the spectrum of R_n [4, 5, 18, 19, 7, 6].

Much less is known about *singular* values; Clément and Steinerberger [9] say so explicitly, and treat only the largest. The idea of routing the Riemann hypothesis through a smallest singular value appears to originate with Bordellès and Cloitre [3], who study an integer matrix Γ_n with $\det \Gamma_n = n! \sum_{k \leq n} \mu(k)/k$ and deduce from an LU decomposition that a lower bound $\sigma_{\min}(U_n) \gg_\varepsilon n^{-1/2-\varepsilon}$ would imply RH; Cheon and Kim [8] give a criterion of the same shape for the triangular factor of a Mertens equimodular matrix, and ask which matrices satisfy it. In both cases the criterion is a *sufficient condition* awaiting verification, and in neither is the relevant singular value determined. The present paper determines the singular values of one family at both ends, and answers the Cheon–Kim question for it.

1.1 The matrix

The following much sparser $(0, 1)$ matrix with the same determinant was introduced in [14, §1]. Let $P(i)$ denote the largest prime factor of $i > 1$, and set

$$\mathcal{A}(i, j) := \begin{cases} 1 & \text{if } i = j, \\ 1 & \text{if } i \text{ is squarefree and } i/j = P(i), \\ 0 & \text{otherwise,} \end{cases} \quad \mathcal{R}_n := \mathcal{A}_n + e_1(\mathbf{1} - e_1)^\top, \quad (2)$$

so that $\mathcal{R}_n$ agrees with $\mathcal{A}_n$ except that its first row is all ones. Then $\mathcal{A}_n$ is unit lower triangular, $\det \mathcal{R}_n = M(n)$, and $\mathcal{R}_n$ has about $2.61n$ nonzero entries: one per diagonal, one per row of the all-ones row, and one for each squarefree $i \leq n$, the last contributing $6n/\pi^2 + O(\sqrt{n})$. By contrast R_n has $\sum_{i \leq n} d(i) = n \log n + (2\gamma - 1)n + O(n^\theta)$ nonzeros. The two dominant eigenvalues of $\mathcal{R}_n$ are

$$r_\pm = 1 \pm \sqrt{\pi(n)} + \frac{1}{2} \frac{\pi'_2(n)}{\pi(n)} + O\left(\frac{\log \log^2 n}{\sqrt{n}/\log n}\right) \quad (3)$$

by [14, Thm. 1], where $\pi'_k(n)$ counts squarefree integers below n with exactly k prime factors.

The reason $\sqrt{\pi(n)}$ appears in (3), rather than the $\sqrt{n}$ of the Redheffer matrix [5, Thm. 1], is worth stating because the same mechanism governs $\sigma_{\max}$ below. Writing $\mathcal{A} = I + S$, the matrix S is the adjacency matrix of a *forest*: each squarefree i has the unique parent $i/P(i)$. The children of the root 1 are exactly the primes, so the hub degree is $\pi(n)$; in R_n the corresponding hub degree is n .

1.2 Results

Throughout, $P^-(i)$ is the smallest prime factor of $i > 1$ with $P^-(1) = \infty$, and

$$M(x, y) := \sum_{\substack{d \leq x \\ P^-(d) > y}} \mu(d) \quad (4)$$

is the Möbius function summed over *y-rough* integers. We write $\alpha = \log x / \log y$ and let ρ denote Dickman's function, defined by $\rho(\alpha) = 1$ on $0 < \alpha \leq 1$ and $\rho(\alpha) = 1 - \int_1^\alpha \rho(t-1) dt/t$ for $\alpha > 1$.

Set $\mu_n := (\mu(1), \dots, \mu(n))^\top$ and

$$w := \mathcal{A}_n^{-\top}(\mathbf{1} - e_1). \quad (5)$$

The first of these is the first column of $\mathcal{A}_n^{-1}$; the vector w collects the *other* columns, and it is the object that was missing from the analysis in [15].

Our first result identifies w completely.

Theorem 1 (closed form). *For $1 \leq j \leq n$,*

$$w_j = \begin{cases} 1 & \text{if } j \text{ is not squarefree,} \\ M(n/j, P(j)) & \text{if } j \text{ is squarefree,} \end{cases}$$

with the convention $P(1) := 1$, so that $M(n/1, P(1)) = M(n)$, and with the value at $j = 1$ diminished by 1. Consequently

$$\|w\|^2 = \#\{j \leq n : j \text{ not squarefree}\} + (M(n) - 1)^2 + \sum_{\substack{2 \leq j \leq n \\ j \text{ squarefree}}} M(n/j, P(j))^2.$$

So an inverse-matrix vector for $\mathcal{R}_n$ is, entry by entry, a rough-Möbius sum evaluated along the hyperbola $x = n/j$, $y = P(j)$. Alladi's asymptotic for $M(x, y)$ then controls $\|w\|$:

Theorem 2 (norm). $\|w\| = n^{1+o(1)}$, *unconditionally.*

Combining with an exact Sherman–Morrison factorisation gives the main result.

Theorem 3 (smallest singular value). *Unconditionally,*

$$|M(n)| \cdot n^{-3/2+o(1)} \leq \sigma_{\min}(\mathcal{R}_n) \leq |M(n)| \cdot n^{-4/3+o(1)},$$

the upper bound needing nothing beyond the prime number theorem (Remark 13). If moreover $|M(n)|/\|w\| \rightarrow 0$, the two collapse to $\sigma_{\min}(\mathcal{R}_n) = |M(n)| \cdot n^{-3/2+o(1)}$. In either case

$$RH \iff \sigma_{\min}(\mathcal{R}_n) \ll_\varepsilon n^{-1+\varepsilon}.$$

The equivalence is two-sided and unconditional: “ $\Leftarrow$ ” is the unconditional lower bound rearranged, and “ $\Rightarrow$ ” may assume RH, which supplies $|M(n)|/\|w\| \rightarrow 0$ and hence the exact asymptotic. The hypothesis $|M(n)|/\|w\| \rightarrow 0$ is not known unconditionally: the best available bounds, $\|w\| \gg_\varepsilon n^{1-\varepsilon}$ from Theorem 2 and $|M(n)| \ll n \exp(-c(\log n)^{3/5}(\log \log n)^{-1/5})$, do not cross.

We stress what is and is not content here. That $\det \mathcal{R}_n = M(n)$ is [13, 14], so “RH $\iff |\det \mathcal{R}_n| \ll n^{1/2+\varepsilon}$ ” is immediate and carries nothing new. The content of Theorem 3 is the conversion factor $\|\mu_n\| \|w\| = n^{3/2+o(1)}$, that is, Theorem 2: without it the equivalence is unavailable in either direction at the stated exponent.

At the other extreme, and for the unmodified factor:

Theorem 4 (the other extreme). $\sigma_{\max}(\mathcal{R}_n) = \sqrt{n}(1 + O(1/n))$, and $\sigma_{\max}(\mathcal{R}_n)/|\lambda_{\max}(\mathcal{R}_n)| \asymp \sqrt{\log n}$. Moreover $\|\mathcal{A}_n^{-1}\| = n^{1/2+o(1)}$ and hence $\sigma_{\min}(\mathcal{A}_n) = n^{-1/2+o(1)}$.

The exponent in the second assertion is sharp, but the $o(1)$ is an artefact of the method: the terminating Neumann series bounds $\|\mathcal{A}_n^{-1}\|$ by $\sum_r \sqrt{\pi'_r(n)}$, and that sum exceeds $\sqrt{n}$ by a factor 1.55, 1.62, 1.67, 1.72, 1.76 at $n = 10^4, \dots, 10^8$ — monotone increasing, so the argument cannot deliver $\asymp \sqrt{n}$ however carefully it is run. Numerically $\|\mathcal{A}_n^{-1}\|/\sqrt{n}$ is flat at 0.8499 out to $n = 10^7$, and we expect $\|\mathcal{A}_n^{-1}\| \asymp \sqrt{n}$; we do not prove it.

The last assertion answers a question of [8] for this family; see Remark 15.

1.3 Relation to earlier work

In [15] it is shown that for unit lower-triangular L , with B obtained by deleting the first row and l the first column of L^{-1} , one has $\det BB^\top = \|l\|_2^2$; for the matrices considered here $l = \mu_n$ and $\sqrt{\det BB^\top} = \sqrt{6n}/\pi + O(1)$. The same paper observes, informally, that

the Riemann hypothesis is an assertion that the constant $\mathbf{1}^\top$ is, for all n , almost in the span of the rows of B .

That distance is $|\det \mathcal{R}_n|/\sqrt{\det BB^\top} = |M(n)|/\|\mu_n\| \asymp |M(n)|n^{-1/2}$. Theorem 3 completes the picture quantitatively:

$$\begin{aligned} & \underbrace{\text{dist}(\mathbf{1}^\top, \text{rowspan } B)}_{= |M(n)|/\|\mu_n\| \asymp |M(n)|n^{-1/2}} \quad \text{versus} \quad \underbrace{\sigma_{\min}(\mathcal{R}_n)}_{= |M(n)|/(\|\mu_n\|\|w\|) = |M(n)|n^{-3/2+o(1)}} . \\ & \hspace{10cm} (6) \end{aligned}$$

The first is the distance from one particular vector to a hyperplane; the second is the minimum over all unit directions. They differ by $\|w\| = n^{1+o(1)}$: the matrix is a full factor of $n^{1+o(1)}$ closer to singular than that single direction reveals, and quantifying the discrepancy is exactly what requires Alladi’s estimates.

1.4 What is cited and what is proved

The analytic engine is not ours. Theorem 5 below is [1, Thm. 1], together with the contrast with $\Psi(x, y)$ that he draws on p. 87 and that we record in Remark 6; [2] is its companion. What is

established here is the linear algebra: Theorem 1, which identifies w with Alladi's sums, and the consequences 2–4. All numerical claims are reproduced by the scripts in `code/`; the two independent derivations of Theorem 5 obtained while this work was in progress, and the prior-art audit, are recorded in `audit/reports/`.

2 Notation and cited results

For an arithmetic function f we write $A(f)_n$ for the $n \times n$ matrix with (i, j) entry $f(i/j)$ when $j \mid i$ and 0 otherwise. All matrices below are real. We use $\|\cdot\|$ for the spectral norm and $\sigma_1 \geq \dots \geq \sigma_n$ for singular values, so $\sigma_{\min} = \sigma_n$.

Recall $\mathcal{A} = I + S$ from (2), where

$$S(i, j) = 1 \iff i \text{ squarefree, } i \geq 2, j = i/P(i). \quad (7)$$

Each row of S has at most one nonzero entry, so S is the adjacency matrix of a forest in which every squarefree $i \geq 2$ has the single parent $i/P(i)$.

We use the following two results without proof.

Theorem 5 (Alladi [1, Thm. 1]). *If $y \geq x$ then $M(x, y) = 1$. If $y = x^{1/\alpha}$ then*

$$M(x, y) = \frac{x \rho'(\alpha)}{\log y} + \frac{y}{\log y} + O\left(\frac{x \alpha^2}{\log^2 y}\right),$$

uniformly for $2 \leq y < x$.

Differentiating the defining relation for ρ gives $\alpha \rho'(\alpha) = -\rho(\alpha - 1)$, so Theorem 5 may be written in the equivalent form

$$M(x, x^{1/\alpha}) = -\frac{x}{\log x} \rho(\alpha - 1) (1 + O_\alpha(1/\log y)), \quad (8)$$

which is the shape used below. Note $\rho > 0$ everywhere, so $M(x, y)$ is negative and bounded away from 0 relative to $x/\log x$ for each fixed α .

Remark 6. Alladi states the contrast with the smooth-number count $\Psi(x, y) \sim x\rho(\alpha)$: the same sum weighted by μ produces Dickman's *derivative* ρ' in place of ρ itself [1, p. 87]. It is worth adding, though it is not Alladi's remark, that the unweighted rough count $\Phi(x, y) = \#\{d \leq x : P^-(d) > y\} \sim x\omega(\alpha)/\log y$ involves Buchstab's ω ; so the μ -weighting replaces ω by ρ' .

Theorem 7 ([15, Thms. 1 and 3]). *Let L be unit lower triangular, let B be L with its first row deleted, and let l be the first column of L^{-1} . Then $\det BB^\top = \|l\|_2^2$. For $L = \mathcal{A}_n$ one has $l = \mu_n$ and $\|\mu_n\|^2 = \sum_{i \leq n} |\mu(i)| = 6n/\pi^2 + O(\sqrt{n})$.*

3 The inverse, and the vector w

Proof of Theorem 1. Since S is nilpotent, $\mathcal{A}^{-1} = \sum_{r \geq 0} (-S)^r$, and $(S^r)_{kj}$ counts paths of length r from k to j in the parent forest. Each vertex has at most one parent, so there is at most one such path, and it exists precisely when k is squarefree and j is obtained from k by repeatedly deleting largest prime factors. Writing k squarefree with primes $p_1 < \dots < p_r$, the vertices reachable from k are exactly the initial products $p_1 \dots p_m$, $0 \leq m \leq r$. Hence $(\mathcal{A}^{-1})_{kj} \neq 0$ if and only if k is squarefree, $j \mid k$, and every prime factor of k/j exceeds $P(j)$ — which forces j squarefree as well — and in that case $(\mathcal{A}^{-1})_{kj} = (-1)^{\omega(k/j)} = \mu(k/j)$.

Now $w_j = \sum_{k \geq 2} (\mathcal{A}^{-1})_{kj}$. If j is not squarefree the only surviving term is the diagonal one, giving $w_j = 1$. If j is squarefree, substitute $d = k/j$: the conditions become $d \leq n/j$, d squarefree, and $P^-(d) > P(j)$, whence $w_j = \sum_{d \leq n/j, P^-(d) > P(j)} \mu(d) = M(n/j, P(j))$; for $j = 1$ the constraint $k \geq 2$ removes the term $d = 1$. $\square$

Two consequences are worth isolating, because they make part of the sum in Theorem 1 completely explicit. Write $x = n/j$ and $y = P(j)$.

Corollary 8. *For squarefree j : if $y \geq x$ then $w_j = 1$; if $\sqrt{x} \leq y < x$ then $w_j = 1 - (\pi(x) - \pi(y))$.*

Proof. If $y \geq x$ the only $d \leq x$ with $P^-(d) > y$ is $d = 1$. If $\sqrt{x} \leq y < x$ then such a d is either 1 or a prime in $(y, x]$, since a composite y -rough number is at least $y^2 \geq x$. $\square$

The following two identities for $M(x, y)$ are standard and are used below; both are immediate from unique factorisation.

Lemma 9. $M(x, y) = M(x) + \sum_{p \leq y} M(x/p, p)$, equivalently $M(x, y) = 1 - \sum_{y < p \leq x} M(x/p, p)$; and $M(x, y) = \sum_{a \leq x, a \text{ } y\text{-smooth}} M(x/a)$, the sum being over all y -smooth a , not merely squarefree ones.

4 The norm of w

Proof of Theorem 2. Upper bound. Trivially $|M(x, y)| \leq x$, so by Theorem 1 $\|w\|^2 \leq \sum_{j \leq n} (n/j)^2 \leq \zeta(2)n^2$, giving $\|w\| \leq (\pi/\sqrt{6})n$.

Lower bound. Fix $\alpha > 1$ and let $j = p$ run over the primes in $(n^{1/(\alpha+1)}, 2n^{1/(\alpha+1)}]$. For such p we have $x = n/p$, $y = p$ and $\log x / \log y = \alpha + O(1/\log n)$, so Theorem 5 applies at fixed α : its main term is $x\rho'(\alpha)/\log y \asymp_\alpha x/\log y$ and dominates the error $O(x\alpha^2/\log^2 y)$ by a factor $\log y$. Since $\rho > 0$, $|w_p| = |M(n/p, p)| \gg_\alpha n^{\alpha/(\alpha+1)}/\log n$. By the prime number theorem there are $\gg n^{1/(\alpha+1)}/\log n$ such primes, and by Theorem 1 the quantity $\|w\|^2$ is a sum of squares, so *any* subfamily bounds it below:

$$\|w\|^2 \gg_\alpha \frac{n^{1/(\alpha+1)}}{\log n} \cdot \frac{n^{2\alpha/(\alpha+1)}}{\log^2 n} = \frac{n^{(2\alpha+1)/(\alpha+1)}}{\log^3 n}, \quad \text{i.e.} \quad \|w\| \gg_\alpha \frac{n^{1-\frac{1}{2\alpha+2}}}{\log^{3/2} n}.$$

Letting $\alpha \rightarrow \infty$ gives $\|w\| \gg_\varepsilon n^{1-\varepsilon}$ for every $\varepsilon > 0$. $\square$

Remark 10 (the refined shape, open). Both a saddle-point analysis and the numerics of §7 indicate $\|w\| = n \exp(-(1+o(1))\sqrt{\log n \log \log n})$, the maximising subfamily sitting at $\alpha \asymp \sqrt{\log n / \log \log n}$. This is *not* established. Theorem 5 does not reach that range: as Alladi notes [1, p. 87], since $\rho'(\alpha) = -\exp\{-\alpha \log \alpha - \alpha \log \log \alpha + O(\alpha)\}$ for $\alpha > 3$, “the main terms of Theorem 1 are smaller than the error term when α is large”. His Theorem 2 covers large α but only as an upper bound with decay $\exp\{-(\alpha/2) \log \alpha\}$, half the true exponent. Determining the constant therefore requires an improvement of the 1982 estimates, and would sharpen the $o(1)$ in Theorems 2 and 3.

5 The smallest singular value

Proposition 11. $\mathcal{R}_n^{-1} = \mathcal{A}_n^{-1} - \mu_n w^\top / M(n)$ whenever $M(n) \neq 0$. If moreover $|M(n)|/\|w\| \rightarrow 0$, then

$$\sigma_{\min}(\mathcal{R}_n) \cdot \|\mu_n\| \cdot \|w\| = |M(n)| (1 + o(1)). \quad (9)$$

Proof. Sherman–Morrison applied to $\mathcal{R}_n = \mathcal{A}_n + e_1(\mathbf{1} - e_1)^\top$ gives the stated inverse, using $\mathcal{A}_n^{-1}e_1 = \mu_n$ and $(\mathbf{1} - e_1)^\top \mathcal{A}_n^{-1} = w^\top$, together with $1 + (\mathbf{1} - e_1)^\top \mathcal{A}_n^{-1}e_1 = M(n)$, which is $\det \mathcal{R}_n = M(n)$. The rank-one term has norm $\|\mu_n\| \|w\| / |M(n)|$, while $\|\mathcal{A}_n^{-1}\| = n^{1/2+o(1)}$ by Theorem 4. Their ratio is $(\|\mu_n\| / \|\mathcal{A}_n^{-1}\|) \cdot \|w\| / |M(n)|$, and the first factor is bounded below by a positive constant since $\|\mathcal{A}_n^{-1}\| \geq \|\mu_n\|$; so the rank-one term dominates precisely when $\|w\|/|M(n)| \rightarrow \infty$, which is the hypothesis. Then $\sigma_{\min}(\mathcal{R}_n) = 1/\|\mathcal{R}_n^{-1}\|$ gives (9). $\square$

The hypothesis is harmless in the application — RH supplies it at once — but it is not known unconditionally, since the best lower bound for $\|w\|$ and the best upper bound for $|M(n)|$ do not cross. This is the one place where the chain is conditional, and it is why Theorem 3 is stated as a bracket.

Proof of Theorem 3. By Theorem 7, $\|\mu_n\| = \sqrt{6n/\pi^2} (1 + o(1)) = n^{1/2+o(1)}$, and by Theorem 2, $\|w\| = n^{1+o(1)}$.

Lower bound. The elementary bounds $\|w\| \leq (\pi/\sqrt{6})n$ and $\|\mu_n\| \leq \sqrt{n}$, with $\sigma_{\min}(\mathcal{R}_n) \geq |M(n)|/(\|\mu_n\| \|w\|)$ from the rank-one representation, give $\sigma_{\min}(\mathcal{R}_n) \geq |M(n)|n^{-3/2+o(1)}$ unconditionally. *Upper bound.* Remark 13 gives $\sigma_{\min}(\mathcal{R}_n) \ll |M(n)|n^{-4/3+o(1)}$ from the prime number theorem alone. If $|M(n)|/\|w\| \rightarrow 0$, Proposition 11 applies and substituting the two norms into (9) collapses the bracket to $\sigma_{\min}(\mathcal{R}_n) = |M(n)|n^{-3/2+o(1)}$.

For the equivalence: if RH holds then $M(n) \ll n^{1/2+\varepsilon}$, which forces $|M(n)|/\|w\| \rightarrow 0$ by Theorem 2, so the exact asymptotic is available and $\sigma_{\min} \ll n^{-1+2\varepsilon}$. Conversely, $\sigma_{\min} \ll_\varepsilon n^{-1+\varepsilon}$ combined with the unconditional lower bound forces $M(n) \ll n^{1/2+\varepsilon}$, which is (1). Note that the reverse direction uses only the elementary upper bounds on $\|w\|$ and $\|\mu_n\|$; the forward direction is what Theorem 2 buys. $\square$

Remark 12. $\mathcal{R}_n$ is singular exactly when $M(n) = 0$, which occurs infinitely often, so $\sigma_{\min}(\mathcal{R}_n) = 0$ for infinitely many n and no nontrivial unconditional lower bound is available. Only upper bounds are meaningful, which is why test vectors and projections are the natural tool; see Remark 13.

Remark 13 (a hierarchy of cheaper bounds). For any orthogonal projection Q one has $\|\mathcal{R}_n^{-1}\| \geq \|\mathcal{R}_n^{-1}Q\|$, and different choices of Q recover successively sharper bounds at successively higher cost. Taking the single test vector $\mu_n/\|\mu_n\|$ gives $\sigma_{\min} \leq |M(n)|/\|\mu_n\|$, which is (6) and is weaker than the truth by exactly $\|w\|$. Taking Q of rank one at an index $j = p$ with $p \approx n^{1/3}$ — where Corollary 8 applies and w_p is an *exact* difference of prime counts — already gives $\sigma_{\min} \ll |M(n)|n^{-7/6+o(1)}$ using only the prime number theorem. Summing Corollary 8 over the whole range $n^{1/3} \leq p \leq n^{1/2}$ gives $\sigma_{\min} \ll |M(n)|n^{-4/3+o(1)}$, still with no input beyond PNT. Only the last step from $n^{-4/3}$ to $n^{-3/2}$ requires Theorem 5. In particular, combining the third bound with the unconditional $M(n) \ll n \exp(-c\sqrt{\log n})$ yields the clean estimate $\sigma_{\min}(\mathcal{R}_n) \ll n^{-1/3+o(1)}$ by elementary means.

6 The largest singular value, and the unmodified factor

Proof of Theorem 4. Largest singular value. The first row of $\mathcal{R}_n$ has norm $\sqrt{n}$, so $\sigma_{\max} \geq \sqrt{n}$. For the upper bound write $\mathcal{R}_n = e_1 \mathbf{1}^\top + (I - e_1 e_1^\top) + S$. Each row of S has at most one nonzero entry and the column sums of S are the numbers of children, the largest being $\pi(n)$ at the root; the Schur test therefore gives $\|S\| \leq \sqrt{1 \cdot \pi(n)} = \sqrt{\pi(n)}$, and the star at the root shows this is an equality. The triangle inequality gives only $\sigma_{\max} \leq \sqrt{n} + 1 + \sqrt{\pi(n)} = \sqrt{n}(1 + O(1/\sqrt{\log n}))$, which is weaker than claimed; for the stated error term argue on the Gram matrix instead. Write

$$\mathcal{R}_n \mathcal{R}_n^\top = \begin{pmatrix} n & b^\top \\ b & C \end{pmatrix},$$

the first row of $\mathcal{R}_n$ being $\mathbf{1}^\top$. For $k \geq 2$, b_k is the number of nonzero entries in row k , which is 2 if k is squarefree and 1 otherwise; hence

$$\|b\|^2 = 4(Q(n) - 1) + (n - Q(n)) = 3Q(n) + n - 4 \leq 4n,$$

where $Q(n) = \#\{k \leq n : k \text{ squarefree}\}$. Also $C = BB^\top$ with B the last $n - 1$ rows of $\mathcal{R}_n$, so $\|C\| \leq (1 + \sqrt{\pi(n)})^2 = o(n)$. If $\lambda = \lambda_{\max}(\mathcal{R}_n \mathcal{R}_n^\top) \geq n$ then $\lambda > \|C\|$ for large n , and the Schur complement identity for the top-left entry gives

$$\lambda = n + b^\top (\lambda I - C)^{-1} b \leq n + \frac{\|b\|^2}{\lambda - \|C\|} = n + O(1).$$

Hence $\sigma_{\max} = \sqrt{n + O(1)} = \sqrt{n}(1 + O(1/n))$. (Numerically $\lambda_{\max}(\mathcal{R}_n \mathcal{R}_n^\top) - n \rightarrow 1 + 18/\pi^2 = 2.8238$, which is $\lim \|b\|^2/n$, so the $O(1)$ is attained and not merely an upper bound.)

Non-normality. By (3), $|\lambda_{\max}(\mathcal{R}_n)| = (1 + o(1))\sqrt{\pi(n)}$, and $\sqrt{n/\pi(n)} \asymp \sqrt{\log n}$.

The unmodified factor. Let $\omega_{\max}(n)$ be the largest number of prime factors of a squarefree integer at most n , so $\omega_{\max}(n) \sim \log n / \log \log n$. Each application of S deletes one prime factor, hence $S^{\omega_{\max}(n)+1} = 0$ and the Neumann series $\mathcal{A}_n^{-1} = \sum_{r \leq \omega_{\max}(n)} (-S)^r$ terminates. Row sums of S^r are at most 1 and its column sums are $\pi'_r(n)$, the number of squarefree $k \leq n$ with $\omega(k) = r$; so $\|S^r\| \leq \sqrt{\pi'_r(n)}$ and $\|\mathcal{A}_n^{-1}\| \leq \sum_r \sqrt{\pi'_r(n)} = n^{1/2+o(1)}$. Conversely $\|\mathcal{A}_n^{-1}\| \geq \|\mathcal{A}_n^{-1}e_1\| = \|\mu_n\| \asymp \sqrt{n}$ by Theorem 7. Finally $\sigma_{\min}(\mathcal{A}_n) = 1/\|\mathcal{A}_n^{-1}\|$. $\square$

Remark 14 (Bordellès–Cloitre). The strategy of bounding a smallest singular value from below in order to reach PNT or RH is due to [3]. There U_n is the upper triangular factor of Γ_n , with $U_n^{-1} = (\theta_{ij})$ given explicitly in terms of iterated Möbius functions, and Corollary 2.7 of that paper states $|\sum_{k=i}^n \mu_i(k)/k| \leq 1/(n\sigma_n)$, so that $\sigma_n \gg_\varepsilon n^{-1+\varepsilon}$ suffices for PNT and $\sigma_n \gg_\varepsilon n^{-1/2-\varepsilon}$ suffices for RH. The authors observe that the general bounds available for smallest singular values of triangular matrices — for instance $\sigma_n \geq \min_i |a_{ii}|/2^{n-1}$ under a diagonal dominance hypothesis — are exponentially weak and “still very far from the PNT”.

Theorem 4 is a case where the arithmetic structure does determine the order exactly: for the triangular factor $\mathcal{A}_n$ considered here, $\sigma_{\min}(\mathcal{A}_n) \asymp n^{-1/2}$, obtained not from a general matrix inequality but from the fact that the Neumann series for $\mathcal{A}_n^{-1}$ terminates after $\sim \log n / \log \log n$ terms. We note in passing that this is precisely the exponent appearing in the RH threshold of [3], though for a different matrix, so the two do not combine.

Remark 15 (a question of Cheon and Kim). For a Mertens equimodular matrix $L_n + uv^\top$, since $\det L_n = 1$ and $uv^\top$ has singular values $\sqrt{n-1}, 0, \dots, 0$, the determinantal inequality of Li and Mathias [16] gives $|M(n)| \leq 1 + \sqrt{n-1}/\sigma_{\min}(L_n)$; Cheon and Kim [8, Thm. 4.2] deduce that $1 + \sqrt{n-1}/\sigma_{\min}(L_n) = O(n^{1/2+\varepsilon})$ implies the Riemann hypothesis, and ask which matrices satisfy this. Theorem 4 answers the question negatively for the present family: with $\sigma_{\min}(\mathcal{A}_n) = n^{-1/2+o(1)}$ the quantity is $n^{1+o(1)}$, which is not $O(n^{1/2+\varepsilon})$, so the resulting bound $|M(n)| \leq n^{1+o(1)}$ is trivial. Note their criterion concerns the *unmodified* factor and is one-directional, whereas Theorem 3 concerns the modified matrix and is an equivalence.

7 Numerical remarks

All values below are produced by the scripts in `code/`, using only NumPy (`checkB.py` additionally uses `mpmath`): the $O(n \log n)$ quantities by `wnorm.py` and `massprofile.py`, and every singular-value, eigenvalue and nilpotency figure by `spectra.py`, which builds the matrices explicitly and calls a dense SVD. Two independent implementations were used throughout: a dense construction with a direct linear solve, and an $O(n \log n)$ evaluator built from Theorem 1 and Corollary 8. They agree to residual exactly 0 at $n = 500, 1000, 2000$, which is the check on Theorem 1.

n	$\ w\ $	$\ w\ /n$	local exponent
10^5	2289.3	0.022893	—
$3 \cdot 10^5$	5211.8	0.017373	0.7488
10^6	12850.5	0.012851	0.7496
$3 \cdot 10^6$	29519.5	0.009840	0.7570
10^7	73972.3	0.007397	0.7630
$3 \cdot 10^7$	172026.3	0.005734	0.7683

Table 1: The norm $\|w\|$. The local exponent climbs steadily; Theorem 2 asserts it tends to 1.

Verification of (9), computed from the dense matrix: the ratio $\sigma_{\min}(\mathcal{R}_n)\|\mu_n\|/\|w\|/|M(n)|$ equals 0.9896, 1.0038, 1.0047 at $n = 500, 1000, 2000$. Over the grid $n = 200, 225, \dots, 3200$ (excluding the n with $M(n) = 0$, where $\mathcal{R}_n$ is singular) it oscillates within $[0.987, 1.027]$, the extremes occurring at $n = 300$ and $n = 225$; the deviation is bounded and shows no trend, consistent with the $1 + o(1)$ of Proposition 11, but it is not monotone and does not visibly contract over this range. For $\|\mu_n\|$, the ratio $\|\mu_n\|/\sqrt{n}$ equals 0.77971 at $n = 10^5$ and 0.77970 at $n = 3 \cdot 10^5, 10^6, 3 \cdot 10^6, 10^7$, against $\sqrt{6/\pi^2} = 0.779697$: the squarefree count is already at its asymptotic density to five figures across this range.

For Theorem 4: $\sigma_{\max}(\mathcal{R}_n)/\sqrt{n}$ equals 1.00387, 1.00191, 1.00094, 1.00047, 1.00023 at $n = 400, \dots, 6400$, the excess halving at each doubling; $\sigma_{\max}/|\lambda_{\max}|$ equals 1.906, 2.087, 2.262, 2.440, 2.592 against $\sqrt{\log n} = 2.448, \dots, 2.960$; the nilpotency index of S is 5, 5, 5, 6 at $n = 500, \dots, 3000$; $\|\mathcal{A}_n^{-1}\|/\sqrt{n}$ equals 0.853, 0.851, 0.849, 0.850; and $\sigma_{\min}(\mathcal{A}_n)\sqrt{n}$ equals 1.1717, 1.1753, 1.1773, 1.1760. The quantity of Remark 15 equals 427, 851, 1699, 2552 against the required $O(n^{1/2+\epsilon}) = 22, 32, 45, 55$.

For scale, the same non-normality ratio computed for Redheffer's own R_n (built directly, and checked against $\det R_n = M(n)$) equals 1.2243, 1.2475, 1.2686, 1.2875 at $n = 400, 800, 1600, 3200$, against 1.9063, 2.0871, 2.2623, 2.4395 for $\mathcal{R}_n$. Both are far below $\sqrt{n}$, and at these sizes the sparser matrix is the more non-normal of the two; we draw no asymptotic conclusion from four points.

7.1 The problem is severely pre-asymptotic

We record this because computation up to $n = 10^7$ is actively misleading here, and several natural conjectures it suggests are false.

Table 1 makes $\|w\|$ look like a clean power of n : over $10^5 \leq n \leq 10^6$ the ratio $\|w\|/n^{3/4}$ is stable to four significant figures (0.40711, 0.40658, 0.40637). It is not a power. The ratio breaks at $3 \cdot 10^6$ (0.40951) and the local exponent continues to climb past $3/4$, consistent with Theorem 2. Likewise, the normalisation $\|w\|(\log n)^{1.13}/n^{5/6}$ appears stable to under 1% per decade while in fact increasing monotonically at every step.

The mechanism is a *corner*. Writing $\beta = \log j / \log n$, the mass of $\|w\|^2$ concentrates at $\beta = 1/3$, equivalently $\alpha = 2$ — exactly the threshold of Corollary 8, below which the y -rough numbers under x are still only 1 and the primes. Dickman's ρ has a corner at 1, so the relevant integrand has a jump in its derivative there, and the maximiser is *pinned* at $\beta = 1/3$ until $\log n$ exceeds an explicit threshold. The measured mass-mean β is 0.3368, 0.3364, 0.3361, 0.3351 at $n = 10^5, \dots, 10^7$: already drifting, and accelerating. The crossover at which the $\alpha = 3$ subfamily overtakes the $\alpha = 2$ subfamily lies near $n = 10^{12}$, far beyond any feasible computation.

The practical consequence is that the exponent in Theorem 2 cannot be inferred numerically, and any conjecture about $\|w\|$ or $\sigma_{\min}(\mathcal{R}_n)$ based on data below 10^8 should be treated with suspicion.

8 Open problems

1. **The refined shape.** Establish $\|w\| = n \exp(-(1+o(1))\sqrt{\log n \log \log n})$, and thereby replace the $o(1)$ in Theorem 3 by an explicit function. By Remark 10 this requires extending Theorem

5 to $\alpha \asymp \sqrt{\log x / \log \log x}$ — the μ -weighted analogue of Hildebrand’s range for $\Psi(x, y) \sim x\rho(\alpha)$ [12] — or improving the decay in [1, Thm. 2] from $\exp\{-(\alpha/2) \log \alpha\}$ to $\exp\{-\alpha \log \alpha\}$.

2. **Other families.** By Remark 15 the present family fails the Cheon–Kim criterion because its triangular factor is poorly conditioned, $\sigma_{\min}(\mathcal{A}_n) \asymp n^{-1/2}$. Is there a family in which $\sigma_{\min}(L_n) \gg n^{-\varepsilon}$? Theorem 4 shows the obstruction here is that $\|\mathcal{A}_n^{-1}\|$ is already attained by its first column, namely $\|\mu_n\|$; any candidate family must avoid that. The same question for the matrix U_n of [3] appears to be open, and Theorem 4 suggests that a termination argument for the relevant Neumann series, rather than a general triangular-matrix inequality, is the tool to try.
3. **Deeper singular values.** (3) and Theorem 4 describe the extremes. The forest structure suggests that successive shells — squarefree integers with 2, then 3 prime factors — govern the next singular values, with $\pi'_2(n)$ already visible in (3).

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